MILITARY SPECIFICATION

EJECTOR ASSEMBLIES, AIR

1. SCOPE

1.1 Scope. - This specification covers air ejector assemblies to serve surface condensers for Naval Shipboard application.

1.2 Classification. - Air ejector assemblies shall be of the following types as specified (see 6.1):

Type A - For service of main, ship's service or other auxiliary surface condensers (see 3.3.1).
Type B - For service of distilling condensers (see 3.3.2).
Type C - Condensers for service of heating drain system or turbine gland leak-off system (see 3.3.3).

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-857 - Bolts, Nuts and Studs.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application, Tests for.
MIL-D-963 - Drawings, Production, Procedure for Procurement of.
MIL-P-2863 - Packing, Condenser-Tube, Flexible-Metallic (Symbol 1435).
MIL-T-15005 - Tubes, 70-30 and 90-10 Copper Nickel Alloy, Condenser and Heat Exchanger.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).
MIL-R-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15430 - Condensers, Steam, Surface, Naval Shipboard.
MIL-C-15726 - Copper-Nickel Alloy Rods, and Flat Products (Flat Wire, Strip, Sheet, Bar, and Plate).
MIL-E-15809 - Expander, Tube, Condenser and Heat Exchangers.
MIL-B-15939 - Aluminum Bronze Bars, Plates, Rods, Sheets, Strips, Forgings, and Structural and Special Shaped Sections.
MIL-P-17286 - Preservation, Packaging, Packing, and Marking of Propulsion and Auxiliary Steam Turbines and Gears (Includes Associated Repair Parts).
MIL-A-17472 - Asbestos Sheet Compressed (Packing Material).
MIL-C-17728 - Control and Drive Assembly, Tube Expander, Automatic.
MIL-D-18641 - Distillation Units, Water: Steam, Or Flashed Vapor Operated, Or Hot Fresh Water Heated, Low Pressure, Naval Shipboard.
MIL-A-19521 - Anodes, Corrosion Preventive, Zinc; Design of and Installation in Shipboard Condensers and Heat Exchangers.
MIL-V-20065 - Valve, Relief, Naval Shipboard.
MIL-P-20087 - Paint, Heat Resisting.
MIL-R-21252 - Rubber Sheet, Solid, Synthetic, Shipboard Water Evaporator Gasketing.

STANDARDS

MILITARY

MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

FSC 4320

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DRAWINGS

BUREAU OF SHIPS
B214 - Root Connections for Attaching Piping.
5000 - S4800-3000 - Schedule for Piping, Pipe Fittings, Valves and Types of Joints Used in Piping Systems.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

A7 - Steel for Bridges and Buildings.
A27 - Mild-to Medium-Strength Carbon-Steel Castings for General Application.
A53 - Welded and Seamless Steel Pipe.
A106 - Seamless Carbon-Steel Pipe for High-Temperature Service.
A108 - Cold-Finished Carbon-Steel Bars and Shafting.
A167 - Corrosion-Resisting Chromium-Nickel Steel Plate, Sheet and Strip.
A193 - Alloy-Steel Bolting Materials for High-Temperature Service.
A276 - Hot-Rolled and Cold-Finished Corrosion-Resisting Steel Bars.
A285 - Low and Intermediate Tensile Strength Carbon-Steel Plates of Flange and Firebox Qualities (Plates 2 inches and under in thickness).
B42 - Seamless Copper Pipe, Standard Sizes.
B98 - Copper-Silicon Alloy Rod, Bar and Shapes.
B111 - Copper and Copper-Alloy Seamless Condenser Tubes and Ferrule Stock.
B127 - Nickel-Copper Alloy Plate, Sheet and Strip.
B139 - Phosphor Bronze Rod, Bar and Shapes.
B143 - Tin-Bronze and Leaded Tin-Bronze Sand Castings.
B148 - Aluminum Bronze Sand Castings.
B152 - Copper Sheet, Strip, Plate and Rolled Bar.
B164 - Nickel-Copper Alloy Rods and Bars.
B171 - Copper-Alloy Condenser Tube Plates.

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification. - The air ejector assemblies furnished under this specification shall be a product which has been tested and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 Materials. - Materials not definitely specified hereinafter shall be of the quality best suited for the purpose intended.

3.2.1 Threaded parts. —Threaded parts shall conform to Handbook H28. The design shall be such that standard wrenches can be used throughout.

3.2.2 Flanges, valves, piping and fittings shall be in accordance with Drawing 5000-S4800-3000 for the applicable service. Root connections shall be in accordance with Drawing B-214. In lieu of Drawing B-214, nozzles may be fabricated of short lengths of pipe with slip-on flange welded to one end, and pads may be of bar stock or plate of requisite thickness for studs. Nozzles shall be of minimum length possible for through bolts.

3.2.3 Welding and allied processes. —Welding and allied processes shall be in accordance with Standard MIL-STD-278. The stress relieving of the inter and after condenser will not be required provided that the carbon content of carbon steel used in welded assemblies does not exceed 0.35 percent.

3.3 Definition of air ejector assembly types. —

3.3.1 Type A air ejector assembly. —Type A air ejector assembly shall serve a main, a ship's service or other auxiliary surface condenser (types I, II and III of Specification MIL-C-15430).

3.3.1.1 A type A air ejector assembly shall consist of the following alternative combinations:

(a) Two 2-stage air ejector units with a surface type inter and after condenser. This combination shall be provided to serve condensers for surface ships only.

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(b) Two 2-stage air ejector units, of which the first stage are noncondensing, the second stage being designed to handle the motive steam from the first stage as well as the air handled by the unit. A surface type after condenser only will be provided with these air ejector units. This combination may be provided to serve condensers for surface ships and shall be provided to serve condensers for submarines.

3.3.2 Type B air ejector assembly. —A type B air ejector assembly shall serve the distiller condenser or the last stage condenser of a distilling unit as covered in Specification MIL-D-18641.

3.3.2.1 A type B air ejector assembly shall consist of one air ejector unit with a surface type after condenser. The air ejector unit shall be either single stage or multiple stage as required by the characteristics of the distilling unit which it serves. Two stage ejector units shall be a design which does not require an intercondenser. A separate intercondenser may be provided with a three stage air ejector unit to condense the steam from the first stage.

3.3.3 Type C air ejector. —A type C air ejector shall serve the condenser provided for the ship's heating drain system or shall be provided for evacuating the air from the turbine gland leak-off system.

3.3.3.1 A type C air ejector shall consist of a single stage and shall be provided in combination with the ship's heating drain system condenser or as a gland exhaust ejector in combination with a type A air ejector assembly.

3.4 General design. -

3.4.1 Supports for the air ejector assembly shall be designed to secure it adequately against high impact shock (see 3.4.2) and when the ship is under the following conditions of inclination:

Surface ships:

(a) Up to 5 degrees from the normal horizontal position in the fore and aft plane (permanently inclined).

(b) Up to 15 degrees to either side (permanently inclined).

(c) With the ship rolling up to 45 degrees from the vertical to either side or pitching 10 degrees up and down from the normal horizontal plane.

Submarines:

(a) Up to 30 degrees from the normal horizontal position in the fore and aft plane (permanently inclined).

(b) Up to 15 degrees to either side (permanently inclined).

(c) With the ship rolling up to 60 degrees from the vertical to either side or pitching 10 degrees up or down from the normal horizontal plane.

The air ejector assembly shall not be supported by plates or brackets in such manner that the primary means of support is the clamping action of the bolts securing the inter and after condenser shell end flange-tube sheet-water box flange joints. Nozzle flanges or piping shall not be utilized for support of the air ejector assembly or its components. The air ejector units shall be supported from the inter and after condenser shell by supplemental supports as necessary to secure against damage from shock.

3.4.2 Shock resistance. -

3.4.2.1 Complete air ejector assemblies weighing 4500 pounds or less shall withstand the tests specified in 4.2.4.

3.4.2.2 Shock resistant design. -

3.4.2.2.1 For assemblies weighing over 4500 pounds, the sub-bases, feet, structural members and holding down bolts shall be designed for the shock design numbers of figure 1. Calculations shall be submitted upon request of the bureau or agency concerned.

3.4.2.2.2 Unless otherwise specified in the contract or order, bolts designed to be stressed in shear shall be installed in holes no greater than the following sizes:

Nominal bolt diameter (inches)	Maximum diameter of hole (inches)
3/4 and smaller	Nominal bolt diameter +1/32
Larger than 3/4	Nominal bolt diameter +1/16

3.4.2.2.3 Assemblies that are rigidly supported shall not be attached to structures which can deflect relative to each other under shock loadings.

3.4.2.2.4 Shock mounts shall not be used without prior approval of the bureau or agency concerned.

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3.4.3 In order to provide for expansion of the shell (see 3.5.2.9.3 and 3.7.4.3) the foot at one end of the shell shall be designed to allow changes in dimension of the length of the shell. Acceptable methods are:

(a) One foot shall be rigid and shall be designed, together with its foundation bolts, to withstand the whole high impact shock loading in the direction longitudinal of the shell. The other foot shall be designed to flex with changes in dimension of the length of the shell. Both feet shall be solidly bolted to the foundation.

(b) One foot shall be rigid and shall be designed, together with its foundation bolts, to withstand the whole high impact shock load in the direction longitudinal of the shell. This foot shall be solidly bolted to the foundation. The other foot shall be rigid and shall be provided with slotted holes for foundation bolts. Bushings shall be furnished around the foundation bolts and shall be provided with a shoulder of larger diameter than the width of the slotted hole. Clearance between the under surface of the shoulder and a finished or spot faced surface on the top of the foot shall be provided, but shall not exceed 0.005 inch. Clearance between foundation bolt and the bore of the bushing and between the outer diameter of the bushing and the width of the slotted hole shall be in accordance with 3.4.2.2.2. The foundation bolts shall hold the bushings solidly to the foundation.

3.4.4 Air ejector units shall have sufficient capacity to serve the condenser with which they operate over the entire range of the condenser operation (see 6.1). The choice of capacity shall be the responsibility of the condenser or distilling unit manufacturer. However, the capacity of each air ejector unit of a type A assembly (see 3.3.1) shall be not less than shown in table I.

Table I - Capacity of type A condenser.

Maximum steam condensed pounds per hour	Capacity, cubic feet per minute 70 degrees free dry air
Up to 12,500	1.5
12,501 to 25,000	3.0
25,001 to 50,000	4.0
50,001 to 100,000	5.0
100,001 to 250,000	7.5
250,001 to 500,000	10.0
500,001 and up	12.5

3.4.4.1 Air ejector units for type A assemblies shall have sufficient capacity to handle the amounts of free dry air specified in 3.4.4 plus the water vapor to saturate the air. The rated saturation pressure shall be considered as 1-inch of mercury (Hg.) absolute or as 1-inch Hg. less than the absolute pressure at the steam inlet of the condenser served, based on the full power ahead steam condensed and the design inlet circulating water temperature specified in Specification MIL-C-15430, whichever absolute pressure is greater. The saturation temperature shall be considered as 7-1/2°F. less than the steam temperature corresponding to the absolute pressure specified herein.

3.4.5 Air ejector units shall be constructed of the material specified in table II.

Table II - Material for air ejector units

Part	Material	Applicable Document
Steam chest	Cast steel, grade 60-30	ASTM A27
	or steel bars, grades 1008 thru 1035	ASTM A108
Suction Chamber	Gun metal, alloy 1-B	ASTM B143
	or cast steel, grade 60-30	ASTM A27
	or fabricated steel or valve bronze,	ASTM A285 (firebox)
	alloy 2A or aluminum	ASTM B143
	bronze alloy, alloy 9D	ASTM B148
Diffusers	Gun metal, alloy 1-B	ASTM B143
	or valve bronze, alloy 2A	ASTM B143
	or aluminum bronze alloy 9D	ASTM B148
Nozzles	Steel, corrosion resisting, type 303	ASTM A276
Nozzle rings	Steel, corrosion resisting, type 303	ASTM A276
Gaskets (nozzle)	Copper, annealed,	----------
	commercial or steel, sheet, hot-rolled, commercial	----------
Bolts and nuts	----------	MIL-B-857 or ASTM A193, grade B7 (high temperature service)

3.4.6 A steam strainer shall be furnished as part of each ejector steam chest. Strainers shall be made of material conforming to table III. Steam supply valves and piping shall not be furnished with the air ejectors.

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Table III - Material for strainers.

Part	Material	Applicable document
Baskets	Steel, corrosion-resisting, type 304 or nickel-copper-alloy	ASTM A167 ASTM B127
Castings	Cast steel, grade 60-30 or bar, grades 1008 thru 1035	ASTM A27 ASTM A108

Steam flow shall be directed into the inside of the strainer baskets, and convenient bolted cleanout covers shall be provided for removal of impurities. Drilled holes in strainer baskets shall be not more than 1/16 inch in diameter and the ratio of the area through strainer basket holes to the area of the inside diameter of the strainer steam inlet shall be not less than 2.5 to 1.

3.4.7 Air ejector units shall be designed for a single steam nozzle in each stage, and shall operate with steam pressures and temperatures as specified (see 6.1).

3.4.8 Air ejector nozzles. —Nozzles shall be accurately machined to size, with throats highly polished and free from imperfections. Nozzles which are provided with threaded ends for screwing the nozzles into the steam chests shall be provided with flat sections for application of a standard wrench to aid in removal, or other means as approved by the bureau or agency concerned.

3.4.9 The after condenser shall be provided with an air vent of generous size to prevent imposition of excessive back pressure which would affect operation of the air ejectors when the vent is piped to a point remote from the unit. The vent shall also be of adequate size to pass the full steam flow from both air ejectors, in case of failure of circulating water, without the pressure in the inter and after condensers exceeding that for which they are designed.

3.4.10 The intercondenser section of the shell shall be provided with a connection for a compound gage.

3.5 Inter, after and gland exhaust condensers of type A air ejector assemblies (surface ships only). —

3.5.1 In order that lightweight and compactness will be obtained (see 3.3.1.1 (a)), the inter, after and gland condensers shall be combined in one shell. The entire assembly shall be designed so that both nozzles of the air ejector not in operation can be dismantled for cleaning and repairs without interfering with the operation of the other air ejector. Inter and after condensers may be subdivided into separate sections to serve each air ejector or may be furnished in single sections and isolating valves provided at the first stage discharge and second stage suction. Isolating valves shall be low-pressure gage or angle valves, or may be suitable manifold valves incorporated in the design of the unit, provided that no undue restriction to the flow of the air-vapor mixture is incurred. All valves shall be fitted with suitable indicators to show whether such valves are in open or shut position.

3.5.1.1 Whenever interstage isolating valves are installed or any possibility of the building up of steam pressure from the steam supply exists, suitable sentinal relief valves shall be installed on the suction chambers and the inter condenser shell. Relief valves shall conform to type A-1, class C of Specification MIL-V-20065.

3.5.2 Referring to both 3.3.1.1(a) and (b):

3.5.2.1 When a gland exhaust condenser section is required to be furnished as part of the after condenser and a gland exhaust fan (not to be furnished by the air ejector manufacturer) is to be used (see 6.1), the after condenser surface shall be increased sufficiently to take care of the additional steam and air from the gland exhaust. When a gland ejector (type C) is specified (see 6.1), it shall be furnished as part of the air ejector assembly, and the gland condenser section shall be integral with the after condenser but subdivided from the after condenser surface, so that the gland steam is condensed in the gland exhaust section and the air is evacuated therefrom by the gland ejector and discharged into the after condenser.

3.5.2.2 Condensate only shall be used for circulation through the inter and after condensers and gland exhaust condenser on surface ships. Refer to 3.6 for submarines.

3.5.2.3 To reduce the necessity for augmenting cooling water by recirculation or other means at low powers or outputs, the quantity of circulating water required for satisfactory operation shall be kept at a minimum. The friction losses shall be the minimum practicable.

3.5.2.4 Sufficient surface shall be provided in the inter, after and gland exhaust condensers so that both two stage air ejector units may be operated at rated capacity and so that there is no discharge of free steam from the after condenser vent when both two stage air ejector units are operating and the gland exhaust condenser is handling the rated amounts of gland leak-off steam and air. When a gland exhaust air ejector is provided, its motive steam shall also be effectively condensed.

3.5.2.5 Tubes and tube sheets. —

3.5.2.5.1 Inter, after and gland exhaust condenser tubes may be straight or U-bent and shall be 5/8 inch outside diameter, number 18 BWG.

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3.5.2.5.1.1 Straight tubes shall have an ordering tube length in multiples of 6 inches when such length equals or exceeds 4 feet.

3.5.2.5.1.1.1 Straight tubes shall be expanded into the tube sheet holes and belled on a 1/2 inch radius to 3/4 inch outside diameter at the inlet ends and the ends shall be finished flush with the face of the tube sheet. The outlet ends shall be packed with copper foil packing conforming to Specification MIL-P-2863, secured by ferrules.

3.5.2.5.1.2 U-bent tubes shall have a minimum centerline radius at the bend of 15/16 inch.

3.5.2.5.1.2.1 U-bent tubes shall be expanded into the tube sheet holes at each end. The inlet ends shall be belled on a 1/2 inch radius to 3/4 inch outside diameter and the ends shall be finished flush with the face of the tube sheet. The outlet ends may protrude a maximum of 3/16 inch beyond the face of the tube sheet, but in no case shall be within the face of the tube sheet.

3.5.2.5.1.3 Expansion of tubes shall be accomplished by means of an automatic tube expander control of a type in accordance with Specification MIL-C-17728, using a tube expander in accordance with Specification MIL-E-15809. The tube expander shall be so adjusted that expansion of the tube is stopped within 1/8 inch of the inner face of the tube sheet.

3.5.2.5.2 Tube holes in the tube sheets shall be drilled to the diameters indicated or otherwise finished as follows:

(a) Holes for inlet ends of tubes - reamed to 0.626 inch diameter with a tolerance of plus 0.002 inch, inner edge slightly rounded and outer edge rounded on a 1/2 inch radius to a diameter of 3/4 inch.

(b) Holes for outlet ends of expanded tubes shall be reamed to 0.626 inch diameter with a tolerance of plus 0.002 inch, with both inner and outer edges slightly round.

(c) Holes for outlet ends of packed tubes - drilled to 0.635 inch diameter with a tolerance of plus 0.002 inch, counterboard truly concentric to a depth of 3/4 inch and tapped 7/8 inch diameter, 18 threads per inch. The inner edges of holes shall be slightly rounded, the outer edges free from burrs.

3.5.2.6 Tube sheets having all expanded ends shall have a minimum thickness of 3/4 inch. Tube sheets having expanded and packed tubes or all packed tubes shall have a minimum thickness of an inch.

3.5.2.7 The tube holes shall be drilled on a minimum spacing of 13/16 inch center to center, in the case of expanded tubes and on a minimum spacing of 31/32 inch, center to center, in the case of packed tubes.

3.5.2.8 Tube sheets shall be secured to the flanges of the shell by through bolts which shall extend through the water box flanges. Studs or tap bolts shall be used only where it is impracticable to install through bolts. A sufficient number of the bolts shall be collar bolts or stud bolts so fitted as to maintain a positively tight joint between the tube sheets and shell flanges when the water boxes are removed. Bolts shall be renewable without removing the water boxes.

3.5.2.9 Shells. —

3.5.2.9.1 The partition between the inter condenser and the after and gland exhaust condensers and between the sections thereof, if sectionalized, shall be sufficiently wide for an adequate gasket, and particular attention shall be given to the spacing of the bolting, so that no leakage can occur between the inter and after condensers or between the active and idle sections thereof.

3.5.2.9.2 Internal baffles shall be fitted as required to protect adjacent tubes from direct impingement of incoming steam.

3.5.2.9.3 The supporting feet shall be designed to allow for expansion of the shell (see 3.4.3.).

3.5.2.10 The materials used in construction shall be in accordance with table IV.

3.6 After condensers of type A air ejector assemblies (submarines only). —

3.6.1 After condensers of type A air ejector assemblies for submarines shall be constructed in accordance with type IV, class A of Specification MIL-C-15430.

3.6.2 Salt water shall be used for circulation in the after and gland exhaust condensers.

3.7 After condensers of type B air ejector assemblies (except for nuclear propelled ships, both surface and submarines) (see 6.1). –

3.7.1 Material for after condensers of type B air ejector assemblies shall be class 1, 2 or 3 as specified (see 6.1).

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Table IV - Material for inter and after condensers.

Parts	Material	Applicable document
Shells	Steel, plate, marine-boiler or seamless steel pipe or seamless steel pipe, grade A or steel, plate	ASTM A285 (firebox) ASTM A53 [2] ASTM A106 ASTM A7 [2] (open hearth)
Water boxes	Steel, plate, marine-boiler or seamless steel pipe or seamless steel pipe, grade A or steel, plate	ASTM A285 (firebox) ASTM A53 [2] ASTM A106 ASTM A7 [2]
Tube sheets	Bronze, aluminum, composition No. 5 or Naval brass	MIL-B-15939 ASTM B171
Tubes	Admiralty metal	ASTM B111 [3]
Ferrules	Admiralty metal	ASTM B111
Tube packing	Copper foil, grade B	MIL-P-2863
Bolts and nuts in contact with steam or condensate [1]	Bronze, aluminum, composition No. 5, stress relieved or phosphor bronze, alloy A or D or copper-silicon alloy	MIL-B-15939 ASTM B139 ASTM B98
Other bolts and nuts	Steel	MIL-B-857
Gaskets between tube sheets and shell flanges	Canvas, cotton (soaked in red lead), commercial or packing, asbestos, compressed	---------- MIL-A-17472
Gaskets between tube sheets and water box flanges	Packing, synthetic rubber, cloth insertion or packing, asbestos, compressed	U. S. Rubber Co. Style 210 or Garlock Packing Co., Style 19, or equal MIL-A-17472

[1] Form in accordance with Specification MIL-B-857.

[2] Maximum carbon 0.35 percent.

[3] Plus tolerance on outside diameter (OD) 0.000 inch.

3.7.1.1 After condensers shall be constructed of class 1 materials as specified in table V.

3.7.1.2 After condensers shall be constructed of class 2 materials as specified in table V, except that copper-nickel alloy, (ASTM Publication B171 [1] and MIL-T-15005) shall be composition 90-10.

[1] Iron content 1.00-1.75, nominal 1.50 percent.

3.7.1.3 After condensers shall be constructed of class 3 materials as specified in table VI.

3.7.1.4 Separate intercoolers for multiple stage type B air ejectors (see 3.3.2.1) shall be of materials specified in 3.7.1.

3.7.2 Salt water shall be used for circulation in the after condenser.

3.7.3 Tubes and tube sheets. –

3.7.3.1 Tubes shall be 5/8 inch outside diameter, No. 18 BWG. Straight tubes in excess of 4 feet long shall have an ordering length in multiples of 6 inches.

3.7.3.1.1 Tubes may be straight or U-bent.

3.7.3.1.1.1 U-bent tubes shall have a minimum centerline radius of 15/16 inch.

3.7.3.2 The after condenser tubes shall be expanded into the tube holes of the tube sheets at both ends. Expansion of the tubes shall be accomplished by means of an automatic tube expander control of a type in accordance with Specification MIL-C-17728, using tube expander in accordance with Specification MIL-E-15809. The tube expander shall be so adjusted that expansion of the tube is stopped within 1/8 inch of the inner face of the tube sheets.

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Table V - Class I materials for after condensers.

Part	Material	Applicable document
Shells	Copper-nickel alloy, composition 70-30 or 90-10 or copper or bronze, aluminum, composition No. 5 [1] or seamless copper pipe or valve bronze, alloy 2A or gun metal, alloy 1B	MIL-C-15726 ASTM B152 MIL-B-15939 ASTM B42 ASTM B143 ASTM B143
Water boxes	Copper-nickel alloy, composition 70-30 or valve bronze, alloy 2A or gun metal, alloy 1B	MIL-C-15726 ASTM B143 ASTM B143
Tube sheets	Copper-nickel alloy, composition 70-30	ASTM B171[3]
Tubes	Copper-nickel alloy, composition 70-30	MIL-T-15005
Bolts and nuts in contact with salt water, steam, or condensate, or in joints involving salt water tightness [2]	Nickel-copper alloy, class A or bronze, aluminum, composition No. 5, stress relieved or phosphor bronze, or copper-silicon alloy	ASTM B164 MIL-B-15939 ASTM B139 ASTM B98
Other bolts and nuts	Steel	MIL-B-857
Gaskets between tube sheets and shell flanges	Canvas (soaked in red lead) commercial or compressed asbestos	---------- MIL-A-17472
Gaskets between tube sheets and water box flanges	Rubber sheet, synthetic	MIL-R-21252

[1]Shells fabricated of aluminum bronze shall be adequately stress relieved after fabrication to prevent stress corrosion cracking.

[2]Form in accordance with Specification MIL-B-857.

[3]Iron content 0.4-0.7 percent, nominal 0.5 percent.

3.7.3.3 Minimum center to center spacing of the tubes shall be 13/16 inch.

3.7.3.4 Tube holes in the tube sheets shall be drilled and reamed to 0.626 inch diameter with a tolerance of plus 0.002 inch and shall be provided with one groove, 0.025 inch wide by 0.007 inch deep, located 3/8 inch from the outer (circulating water) side of the tube sheet. The outer edges of the holes for the inlet tube ends shall be rounded on a 1/2 inch radius to 3/4 inch outside diameter; those for the outlet tube ends shall be free from burrs. The inner edges of the tube holes shall be slightly rounded.

3.7.3.5 Tube sheets shall have a minimum thickness of 3/4 inch.

3.7.3.6 Tube sheets shall be secured to the flanges of the shell as specified in 3.5.2.8.

3.7.4 Shells. –

3.7.4.1 Internal baffles shall be fitted in the shell as required to protect adjacent tubes from direct impingement of the incoming steam.

3.7.4.2 The shell shall be provided with positive means to allow the free expansion and contraction of the tubes. Such means shall be in the form of an expansion joint in the shell or floating tube sheet construction. If an expansion joint is furnished in the shell, it shall be located outboard of one supporting foot of the after condenser. Provision shall be made as necessary for protection of the expansion joint against transverse high impact shock. In lieu of the expansion joint or floating tube sheet construction, U-bent tubes may be used.

3.7.4.3 The supporting feet shall be designed to allow expansion of the shell (see 3.4.3).

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Table VI - Class 3 materials for after condensers.

Part	Material	Applicable document
Shells	Bronze, aluminum, composition No. 5 [1]	MIL-B-15939
	or seamless copper pipe	ASTM B42
	or valve bronze, alloy 2A	ASTM B143
	or gun metal, alloy 1B	ASTM B143
	or copper	ASTM B152
Water boxes	Bronze, aluminum, composition No. 5 [1]	MIL-B-15939
	or valve bronze, alloy 2A	ASTM B143
	or gun metal, alloy 1B	ASTM B143
Tube sheets	Bronze, aluminum, composition No. 5	MIL-B-15939
Tubes	Copper-nickel alloy, 90-10 composition	MIL-T-15005
Bolts and nuts in contact with salt water, steam or condensate, or in joints involving salt water thickness [2]	Bronze, aluminum, composition No. 5, stress relieved	MIL-B-15939
	or phosphor bronze, alloy A or D	ASTM B139
	or copper-silicon alloy	ASTM B98
Other bolts and nuts	Steel	MIL-B-857
Gaskets between tube sheets and shell flange	Canvas, cotton (soaked in red lead), commercial or packing, asbestos, compressed	MIL-C-17472
Gaskets between tube sheets and water box flanges	Rubber sheet, synthetic	MIL-R-21252

[1] Shells and water boxes fabricated of aluminum bronze shall be adequately stress relieved to prevent stress corrosion-cracking.

[2] Form in accordance with Specification MIL-B-857.

3.7.5 Zinc protectors. —The water boxes shall be provided with zinc protectors in accordance with Specification MIL-A-19521.

3.8 After condensers of type B air ejector assemblies for nuclear propelled ships (both surface and submarines) (see 6.1).

3.8.1 The after condensers shall be provided with double tube sheets at each end. The double tube sheets shall be provided with a space at least 1/2 inch between them, either by machining one face of one or both tube sheets or by use of a spacer ring. The space between tube sheets shall be provided with a vent and drain to atmosphere. The joint between tube sheets or between tube sheet and spacer ring may be welded or gasketed. When floating tube sheet construction is used (see 3.7.4.2), the packing of the glands shall bear against the edges of the inner and outer tube sheets to seal the joints to the shell and the waterbox. The inner and outer tube sheets need not be solidly joined. A gland follower ring shall be provided, and shall contain pipe-tapped tell-tale holes. A test ring with bolting lugs or flange similar to those on the waterbox shall be provided. This will be substituted for the waterbox when hydrostatic test pressure is applied between the tube sheets to establish that the tube end expansions in the outer tube sheet are satisfactory. The tell-tale holes in the gland follower ring will be plugged for this test.

3.8.2 The requirements specified in 3.7.1 and 3.7.2 shall also be applicable to the after condenser of type B air ejector assemblies for nuclear propelled ships (both surface and submarines).

3.8.3 Tubes and tube sheets. —

3.8.3.1 Tubes shall be in accordance with 3.7.3.1, 3.7.3.1.1 and 3.7.3.1.1.1.

3.8.3.2 The tubes shall be expanded in both tube sheets at each end as specified in 3.7.3.2. In expanding the tubes into the inner tube sheets, care shall be taken that there is no abrupt change in contour of the inner tube wall. The tube expander shall be so adjusted that expansion is started and stopped 1/8 inch inside both faces of the inner tube sheets.

3.8.3.3 Minimum center to center spacing of the tubes shall be 13/16 inch.

3.8.3.4 Tube holes in the tube sheets shall be drilled as specified in 3.7.3.4, except the holes in the inner tube sheet shall not be provided with a groove and the edges of the tube holes on both faces of the inner tube sheet shall be slightly rounded.

3.8.3.5 Minimum thickness of the tube sheets shall be: outer - 3/4 inch, inner - 7/8 inch.

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3.8.3.6 Tube sheets shall be secured to the flanges of the shell as specified in 3.5.2.8.

3.8.3.7 The manufacturer shall be responsible for furnishing sufficient tube span between the double tube sheets so that the tubes will not be overstressed due to radial expansion differentials of the double tube sheets. Calculations indicating this shall be submitted with the drawings for approval.

3.8.4 Shells shall be as specified in 3.7.4.

3.8.5 Zinc protectors as specified in 3.7.5 shall be provided.

3.9 Identification plates. - Identification plates, of sheet or cast brass or bronze shall be provided. Identification plate data shall include the following:

(a) Manufacturer's name.

(b) Government contract number.

(c) Bureau or agency stock number (allow 17 spaces).

(d) Date of manufacture.

(e) Test pressure, ejector parts.

(f) Test pressure, shell side.

(g) Test pressure, tube side (surface ships only).

(h) Blank space for Government inspector's stamp.

(i) Blank space for unit number (2 spaces - to be stamped by shipyard).

3.10 Painting. —Air ejector assemblies shall be painted as specified below:

(a) Non-ferrous castings or parts need not be painted.

(b) All external surfaces of ferrous metal parts shall be thoroughly cleaned and coated with two coats of heat resisting paint in accordance with Specification MIL-P-20087 or a commercial paint of corresponding properties.

(c) Interior surfaces of ungalvanized ferrous parts shall not be painted or coated with preservative.

3.11 Repair parts and tools. —

3.11.1 Stock repair parts shall conform to Specification MIL-R-15137. The following repair parts for each air-ejector assembly to be carried on board shall be furnished to each ship:

(a) Nozzles, first stage, one of each size.

(b) Nozzles, second stage, one of each size.

(c) Nozzle rings, if used, one for each stage.

(d) Nozzle gaskets, 100 percent for each stage.

(e) Strainer baskets, 2.

(f) Packing, if used, 10 percent.

(g) Ferrules, if used, 10 percent.

(h) Zincs for type B after condensers - 100 percent.

(i) Relief valve: as required by Specification MIL-V-20065.

3.11.2 Tools. - The following tools shall be furnished on the basis of one set for each size air ejector assembly furnished for each ship:

(a) Special wrenches or tools needed for service maintenance, including one tube expander for expanding tubes in double tube sheet after condensers. Special tools are defined as those tools not listed in the Navy Stock List of General Stores. (Copies of this stock list may be consulted in the office of the Government inspector.)

(b) Test rings (see 3.8.1) when required.

3.12 Drawings. —

3.12.1 Drawings shall be submitted for approval to the bureau or agency concerned within the time specified (see 6.1) and before procurement of material and fabrication are begun, and the air ejector assemblies shall be manufactured in accordance with the drawings as approved.

3.12.2 Drawings shall be furnished in accordance with the requirements of Specification MIL-D-963.

3.12.2.1 Types of drawings. - The following types of drawings are required in lieu of the types listed in MIL-STD-7:

3.12.2.2 An external arrangement drawing, called "Outline Drawing." This drawing shall show all necessary external views of the air ejector assembly, shall include all external dimensions required for reproduction on ship's machinery arrangement drawings, for guidance of the shipyard in designing the foundation structure for the assembly, for installation thereof, and for connection of the assembly to the external piping. The drawing shall show the space required for removal and replacement of tubes in the heat exchanger components, removal of waterboxes, covers of access openings, zinc anodes, location of lifting lugs or

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eyebolts provided for handling of the unit or its components and size of openings therein provided for lifting.

3.12.2.3 A drawing showing complete longitudinal and transverse cross-sectional views of the air ejector assembly which shall be called "Assembly Drawing." This drawing shall show the relationship of all parts, arrangement of tubes in heat exchanger components, method of tube and expansion or packing, baffles, welding of adjacent parts. Liberal use of enlarged views or sections shall be made. If necessary, sub-assembly drawings conforming to the above may be furnished for individual components of the unit. The drawing or drawings shall be such that a thorough understanding of the design and construction of the apparatus may be obtained without reference to related detail drawings.

3.12.2.3.1 The Assembly Drawing shall contain a list of materials showing names of parts with identifying numbers and materials of all parts. The identifying numbers shall also be shown adjacent to the part depicted in the various views, with arrows pointing to the parts.

3.12.2.3.2 The Assembly Drawing shall indicate that the zinc anodes provided in the heat exchanger components using salt water circulation comply with the requirements of Specification MIL-A-19521.

3.12.2.4 Detail drawings of all major parts such as shells, waterboxes, tube sheets, tube supports, baffles, and air ejector elements. The drawings shall be completely dimensioned, with finishes and welding symbols indicated, as required for manufacture.

3.12.2.4.1 Submission of detail drawings, for approval, will not be required if sufficient information is provided on the assembly drawing. Detail drawings will, however, be required as part of the final drawings, in order that the intent of 3.12.2.6(e) may be met.

3.12.2.5 A separate drawing, which shall be a ship's drawing, shall contain a tabulation of the following engineering data:

(a) Pressure and superheat of main supply steam.

(b) Operating pressure and superheat of steam supply to nozzles after passing through pressure-reducing valve.

(c) Steam consumption for each stage.

(d) Total steam consumption for each ejector at the specified steam pressure and superheat conditions.

(e) Air-handling capacity as rated in pounds of dry free air per hour plus associated or entrained vapors at operating absolute pressures and air suction temperatures for at least three separate conditions in the usual operating range.

(f) Cooling surface of inter condensers.

(g) Cooling surface of after and gland steam condensers.

(h) Quantity of cooling water at the expected condensate temperature necessary for satisfactory operation and the resultant temperature rise of cooling water.

(i) Pressure drop through inter and after condensers with the maximum specified quantity of circulating condensate flowing through inter and after condensers.

(j) Maximum permissible temperature of discharge cooling water from inter and after condenser for satisfactory operation, together with the minimum quantity of condensate cooling water necessary.

(k) Weight of assembly.

(l) Hydrostatic test pressures for all pressure parts.[1]

(m) List of repair parts furnished.

(n) Number, wall thickness and ordering length of tubes.

(o) Name and address of the manufacturer of the condenser tube packing.

3.12.2.6 It is the intent that sufficient drawings be provided to:

(a) Provide design information sufficient to assure conformance to the requirements of this specification and compatibility with the ship and ship's systems.

(b) Enable evaluation of the suitability of the design for intended naval use.

(c) Enable evaluation of performance and maintenance capability.

(d) Enable shipyard installation without supplier's assistance.

(e) Enable ship and shore activities to repair and maintain the apparatus.

[1] Except type A after condensers for submarines. In this case reference shall be made to confidential shipbuilder's drawing giving hydrostatic test pressures for all parts subjected to submergence pressure.

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3.13 Technical manuals. —Technical manuals shall be furnished in accordance with type II of Specification MIL-M-15071.

3.14 Workmanship. —Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Qualification tests[1]. — Qualification tests on air ejectors shall be conducted at a laboratory satisfactory to the Bureau of Ships. These tests shall consist of the tests specified in 4.2.1 to 4.2.4, inclusive and the tests specified in 4.3. Air ejectors offered for qualification tests shall be type A. Qualification approval of the type A air ejector shall constitute approval of the type B and C air ejectors.

4.2.1 Standard test conditions. --The standard test conditions shall be as follows:

(a) Steam supply pressure, 150 pounds dry and saturated.

(b) Inlet circulating water temperature to inter and after condenser, 90°F.

(c) Quantity of circulating water to inter and after condenser, 35 pounds of circulating water per pound of steam consumption.

(d) Air-handling capacities and steam consumption obtained for the lower absolute pressure range for saturated air suctions to ejectors at both 80° and 90°F.

4.2.2 Such type A air ejectors under test will not be considered satisfactory if the economy as expressed in pounds of steam per one pound of air exceeds the values at 56 pounds at 1-1/2 inches of Hg. absolute pressure, 10 pounds at 2 inches Hg. absolute pressure, and 7.5 pounds at 2-1/2 inches Hg. absolute pressure, with temperature of the saturated air mixture at 90°F. Based on a saturated air-vapor mixture of 80°F., such type A air ejectors under test will, furthermore, not be considered satisfactory if the economy as expressed in pounds of steam per one pound of air exceeds the values of 11.1 pounds at 1-1/2 inches Hg. absolute pressure, 7.1 pounds at 2 inches Hg. absolute pressure, and 6.3 pounds at 2-1/2 inches Hg. absolute pressure.

[1] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.2 and 6.3).

4.2.3. The air ejectors shall further have a characteristic performance curve such that the capacity for air removal at an absolute pressure at the first-stage suction of 2, 2-1/2, 3 or 4 inches Hg. shall be not less than 1.10, 1.35, 1.45, or 1.60 times the capacity for air removal at an absolute pressure of 1-1/2 inches Hg. at the first-stage suction, with the temperature of the saturated air-vapor mixture at 80°F. The capacity for air removal at an absolute pressure at the first stage suction of 2-1/2, 3, or 4 inches Hg. shall be not less than 1.25, 1.35, or 1.55 times the capacity for air removal at an absolute pressure of 2 inches Hg. at the first-stage suction, with the temperature of the saturated air-vapor mixture at 90°F.

4.2.4 Shock tests of complete air ejector assemblies, weighing 4500 pounds or less, shall be performed in accordance with Specification MIL-S-901 with the following modifications:

(a) Under the test procedure for medium weight equipment, the total number of blows shall be increased to nine. The nine blows shall consist of three groups of three blows each. For each group the height of hammer drop and the initial up travel of the anvil table shall be as shown in table I of Specification MIL-S-901.

(1) If feasible within weight limitations of the medium machine, one blow of each group shall be applied parallel to each of three principal axes of the equipment.

(2) If adaptors to permit the above would result in exceeding machine limitations, each group shall comprise the following:

a. The first blow shall be applied with the unit on the standard (horizontal) adaptor

b. The second blow shall be applied with the equipment rotated 45 degree around its principal axis (using a 45 degree adaptor).

c. The third blow shall be applied with the equipment rotated to incline the principal axis 45 degree to the horizontal (using a 45 degree adaptor)

4.3 Acceptance inspection. —

4.3.1 Examination. —Each air ejector assembly offered for delivery under a contract or order shall be examined to determine conformance with this specification regarding design, material, arrangement, construction, dimensions, fits, finish, and workmanship. Parts and subassemblies shall be examined before final assembly for those features which cannot be examined after assembly but are considered important to satisfactory service use.

4.3.2 Performance tests before shipment. — Each air ejector assembly on each contract or order shall be tested for proper operation and performance before shipment.

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4.3.2.1 The tests specified in 4.3.2 shall consist of the following:

(a) Operation of each air ejector unit and determination of capacity (pounds per hour free dry air plus vapor to saturate) versus absolute pressure, inches Hg. from shut-off to at least 2-inches Hg. below rated capacity point. It shall be demonstrated that the air ejector unit is stable in operation throughout the range tested and that the rated capacity is obtained.

(b) Air ejector units for submarines shall be proved capable of starting and operating against the maximum ambient pressure in the hull (see 6.1).

(c) Steam consumption at the specified inlet steam pressure and temperature shall be determined either by weighing the condensate from the inter and after condenser or by calculation from the measured diameter of the nozzle throat.

4.3.2.2 Nozzles furnished as repair parts shall not be tested in the ejectors with which they are furnished, but the nozzles shall be gaged, using the limit gages employed in their manufacture, and other necessary dimensions shall be taken to insure that they are of the same physical dimensions as the service nozzles.

4.3.3 Hydrostatic tests. -

4.3.3.1 The steam chests and steam strainers of air ejectors shall be subjected to a hydrostatic test pressure of one and one-half times the working pressure except in the case of air ejectors for nuclear propelled surface ships and submarines, the test pressure shall be one and one-half times the full generated steam pressure. Other air ejector parts shall be tested hydrostatically to 50 pounds per square inch gage(p.s.i.g.).

4.3.3.2 The inter condenser, after condenser, and gland condenser shell sections shall be separately tested to a pressure of 50 p.s.i.g. and all evidence of leakage in the welds or across the shell partitions between sections shall be corrected before installation of the tubes.

4.3.3.3 After installation of tubes, the following hydrostatic tests shall be applied:

4.3.3.3.1 Type A and B inter, after and gland exhaust condensers for surface ships (nonnuclear propelled) shall have (a) a 50 pound per square inch (p.s.i.) pressure test applied in the shell to demonstrate that there is no leakage from individual tubes either due to defective tubes or imperfect tube end expansion or packing. (b) After this test, the water side shall be subjected to a test pressure of at least twice the design pressure of the ship's condensate system as specified (see 6.1), but under no circumstances shall this test pressure be less than 50 p.s.i.g. Any leakage in water boxes or gasketed joints shall be corrected.

4.3.3.3.2 Type A after and gland exhaust condensers for submarines shall be tested as specified in Specification MIL-C-15430.

4.3.3.3.3 Type B after condensers for nuclear propelled surface ships and submarines shall have (a) a 50 p.s.i.g. test applied in the shell to demonstrate that there are no defective tubes installed. (b) After this test, the spaces between the double tube sheets shall be tested to 50 p.s.i.g. The drains from the spaces between the double tube sheets shall be examined for evidence of leakage between the expanded tube ends and the walls of the tube holes in the inner tube sheets. Any such leakage shall be corrected by minimum possible rerolling of the tubes into the inner tube sheet(s). (c) The pressure shall then be increased to at least twice the maximum designed discharge pressure of the pump supplying cooling water to the after condenser as specified (see 6.1) and the expanded tube ends in the outer tube sheets shall be examined for leakage. Any such leakage shall be corrected by minimum rerolling of the individual tube. (d) The same pressure as in (c) shall then be applied to the salt water side and any leakage in waterboxes or gasketed joints corrected.

4.3.4 Shock test as required by 3.4.2.1

4.3.4.1 Waiver of shock testing will be considered by the bureau or agency concerned for assemblies if they are of similar weight to those which have successfully passed the tests specified in 4.2.4.

4.3.5 Shipboard acceptance tests. - Final acceptance shall be based on performance tests of the air ejector assemblies after installation aboard ship and in connection with the tests of the condensers which they serve.

4.4 Additional inspection. - Where other specifications form a part of this specification, unless otherwise specified herein, sampling, inspection and tests shall be performed as required by the referenced specification.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of shipboard repair parts. -

5.1.1 Air ejector assemblies. -

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5.1.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion and deterioration during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, contract or order number, contractor's name and destination.

5.1.2 Shipboard repair parts and tools. - Repair parts and tools shall be preserved and packaged level A, packed level C and marked levels A and C in accordance with Specification MIL-P-17286. Items not specifically covered therein shall be preserved and packaged in accordance with the guide lines of Specification MIL-P-116. Items shall be individually packaged unless used in sets in quantities greater than one.

5.2. Domestic shipment and storage or overseas shipment requirements. - The requirements and levels of preservation, packaging, packing and marking shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made by a Government activity (see 6.1)).

5.2.1.1 Preservation, packaging, packing and marking. - Repair parts and tools shall be preserved and packaged level A or C, packed level A or B as specified and marked in accordance with Specification MIL-P-17286 and items not specifically covered therein shall be preserved and packaged in accordance with the guide lines of Specification MIL-P-116. Unless otherwise specified items shall be individually packaged unless used in sets in quantities greater than one.

5.2.1.2 Technical manuals. - Technical manuals (see 3.13) shall be packaged and packed for the level specified and marked in accordance with the applicable manual specification.)

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) If type B is required, specify class of material for after condensers (see 3.7.1) and whether for nuclear or non-nuclear propelled ship (see 3.7 and 3.8).
(d) Supply and operating steam pressures and superheat (see 3.4.7), and maximum ambient pressure in the hull (see 4.3.2.1(b), air ejector units for submarines only).
(e) Air removal capacity of each ejector, given in pounds of free dry air per hour, plus associated vapors at designated absolute pressure and temperature (see 3.4.4).
(f) Maximum, normal, and minimum quantities of condensate available for circulating water in inter and after condensers and design pressure of ship's condensate system (see 4.3.3.3.1).
(g) Temperature of condensate circulating water for supply to inter and after condensers.
(h) Allowable pressure drop through inter and after condensers for designated quantity of condensate circulating water.
(i) Specific limitations as to weight, space, foundations, sizes, and relative positions of all connecting flanges, maximum permissible steam consumption, and all other necessary restrictions where any or all of such limitations are essential to the installation in question. This information should be shown on a space diagram.
(j) Whether a gland exhaust fan will be used or whether a gland ejector is to be furnished as part of air ejector assembly (see 3.5.2.1). Amounts of turbine gland leakoff steam and air and vacuum required.
(k) Limiting date for submission of working drawings for approval after award of contract or order (see 3.12.1).
(l) Maximum designed discharge pressure of the pump supplying circulating water to the type B after condenser on nuclear propelled surface ships and submarines (see 4.3.3.3.3 (c).
(m) Preservation, packaging, packing or marking required other than those required by 5.1 (see 5.1 and 5.2).
(n) Level of packaging and packing required for technical manuals (see 5.2.1.2).

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6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL 15465, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification" (see 6.3).

6.3 Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pa.

Notice. —When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Governement thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Bureau of Ships
(Project 4320-N032Sh)

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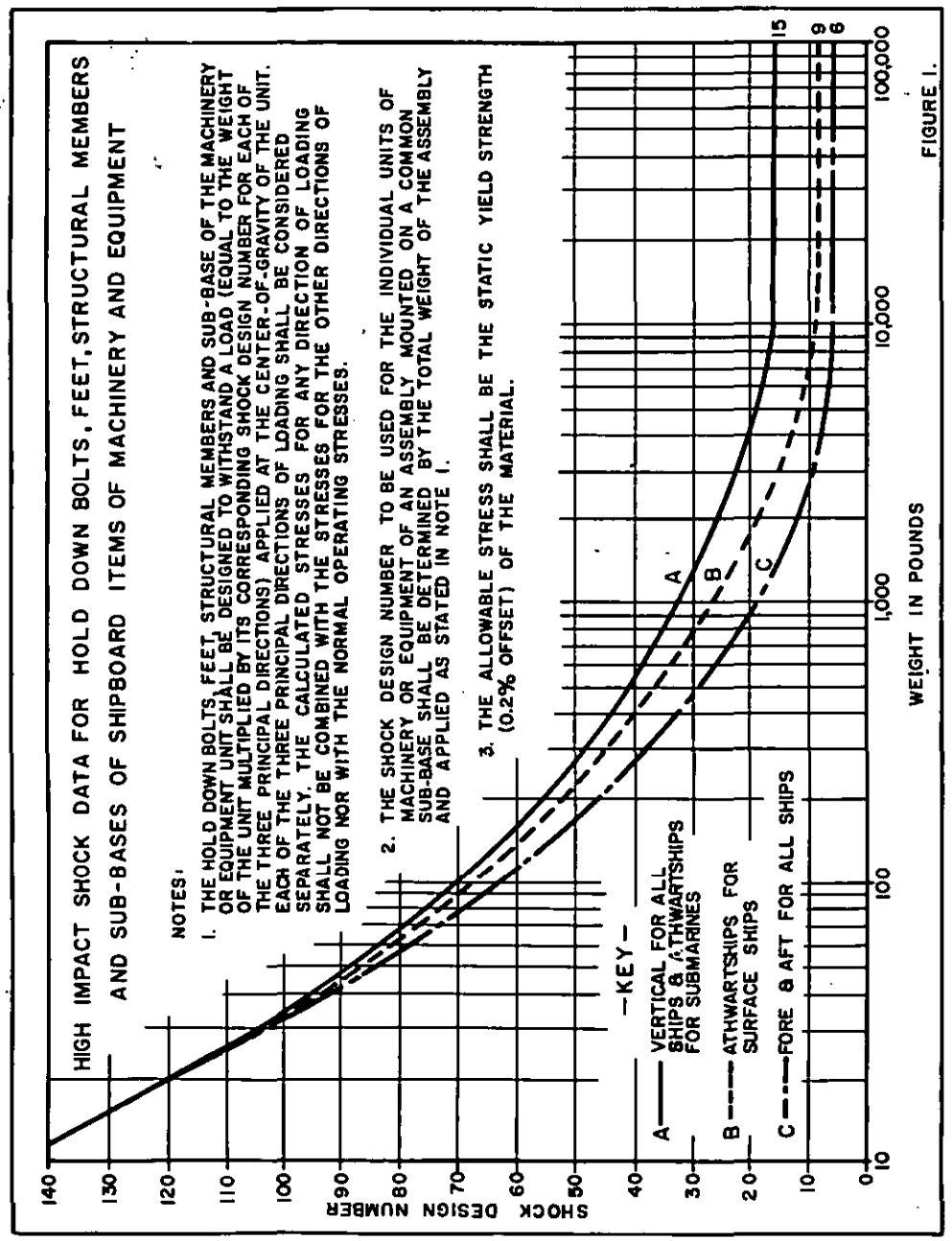


Figure 1 - H. I. Shock Design Data

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PLATE NO. 15419 (BACK)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

C-8279

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